

Q: Find the domain and range of

$$T(x_1, x_2, x_3, x_4) = (x_1, x_2)$$

Solution: The transformation

$$T(x_1, x_2, x_3, x_4) = (x_1, x_2)$$

maps a four-dimensional space to a two-dimensional space by simply discarding the x_3 and x_4 components.

1. Domain: The domain of the transformation consists of all possible inputs (x_1, x_2, x_3, x_4) . Therefore the domain is the four-dimensional space $\mathbb{R}^4$.

2. Range: The range of the transformation consists of all possible outputs (x_1, x_2) . Since the x_3 and x_4 components are disregarded, the possible values for x_1 and x_2 are any real numbers. Therefore the range is the two-dimensional space $\mathbb{R}^2$.